

Arunabha Majumder

Mechatronics Engineer — Product Development, Prototyping & Laboratory Verification

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SUMMARY

Mechatronics engineer who develops energy-efficient electromechanical products from concept to validated prototype. Designed a mechatronic actuation subsystem that improved energy efficiency by more than 25% (Best Paper Award, 2026), with hands-on experience in 3D CAD, prototyping, laboratory testing and verification.

KEY RESULTS

- **>25%** energy-efficiency gain from a new mechatronic subsystem
- **<10%** RMS error on a verified sensing method
- **Concept** → **prototype** → **lab validation** delivered end to end

EXPERIENCE

Research Engineer – Mechatronic Product Development (PhD Project: VIEOX) | 2023 – Present *Aalborg University, Aalborg, Denmark*

- Improved energy efficiency by more than 25%, measured in laboratory testing of an actuated shoulder exoskeleton, by developing a new mechatronic subsystem: a hybrid actuation concept that pairs a parallel spring with a 6 Nm motor (Best Paper Award, IFToMM ISRM 2026).
- Turned design targets into a working functional model, from product concept to detailed design, by creating the 3D CAD design and assemblies in SolidWorks, using FEA for strength, stiffness and weight, and planning, building and evaluating prototypes made from machined and 3D-printed parts.
- Achieved force estimation accurate to below 10% RMS error, verified against a force sensor, by developing a sensorless method for a 5-bar planar parallel robot using variable-stiffness actuators and the robot Jacobian, as part of product verification and performance evaluation.
- Reduced user muscle effort by more than 15%, measured with EMG, by integrating the motor, encoder, force sensors and DAQ with an Assist-as-Needed controller in the embedded control system.
- Delivered a complete laboratory test dataset from a 12-participant study across 3 operating modes and 2 load conditions by running the data collection within a multidisciplinary project team (mechanical, electronics, control, biomechanics).
- Documented results in peer-reviewed publications; mentored and supervised Bachelor's (B.Tech) students (2023 – 2025).

MTech Researcher – Pneumatic Actuator Systems (Master's Thesis) | 2020 – 2022

CSIR-Central Mechanical Engineering Research Institute (CMERI), National Laboratory, Durgapur, India

- Reduced position-tracking error to 0.3–0.78%, outperforming classical PID under varying load, by designing a neural-network-based controller for a non-linear pneumatic actuator.
- Characterised actuator performance through lab testing by fabricating pneumatic artificial muscles in-house and building a test rig with valves, pressure sensors and DAQ.

SKILLS

- **3D CAD:** SolidWorks (CSWA), Fusion 360; quickly transferable to PTC Creo
- **Prototyping & lab testing:** Functional models, test rigs, verification, performance evaluation, DAQ
- **Mechatronics:** Motors, encoders, sensors, pneumatics, embedded control, MATLAB/Simulink, Python
- **Analysis:** FEA, kinematics, control design
- **Languages:** English (C1, fluent), Bengali (Native), Hindi (Conversational)

EDUCATION

Ph.D. Candidate, Mechatronics & Robotics | 2023 – Present

Aalborg University, Department of Materials and Production, Aalborg, Denmark

M.Tech, Mechatronics | 2020 – 2022

AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

B.Tech, Mechanical Engineering | 2015 – 2019

Siddaganga Institute of Technology, Visvesvaraya Technological University (VTU), Tumakuru, India

AWARDS & PUBLICATIONS

- Best Research Paper Award, IFToMM ISRM 2026; Certified SolidWorks Associate (CSWA)
- Two peer-reviewed publications: ISRM 2026 (Springer) and IEEE CONECCT 2022

HANDS-ON PROJECTS

- Rehabilitation exoskeleton (Best Major Project Award and TATA Technologies Innovation Grant, 2019), Go-Kart (2017), Efficycle human-electric hybrid vehicle (2018)

REFERENCES

Available upon request.